Risk and Performance Assessment of a Markowitz Portfolio on CAC 40 Constituents

Numerical Processing of Financial Data

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01/03/2026

Abstract

This report presents the construction and evaluation of a dynamic Markowitz portfolio built from ten constituents of the CAC 40 French stock index. Using one-minute OHLC price data aggregated to a daily frequency over the period September 2022 – March 2023, we estimate the covariance structure and lag-1 autocorrelations of daily returns. A simple AR(1)-based predictor of expected returns feeds into the classical mean–variance optimisation framework to produce daily-rebalanced optimal weights. The portfolio is then assessed through ex-ante and ex-post Sharpe ratios, the upside potential ratio, the maximum drawdown, and an additive decomposition of portfolio volatility across its constituents.

1 Introduction

Portfolio construction and risk management are central problems in quantitative finance. The mean–variance framework introduced by Markowitz (1952) remains a cornerstone of portfolio theory, providing a principled way to trade off expected return against risk.

The objective of this project is to build a dynamic Markowitz portfolio from a subset of CAC 40 stocks, evaluate its performance and risk using standard metrics, and discuss the limitations of the approach. We work with one-minute OHLC (Open–High–Low–Close) price data provided for 40 stocks over the period September 2022 to March 2023. From this high-frequency data, we extract daily closing prices and compute returns at a daily scale.

2 Data Description

2.1 Stock Universe

The full dataset comprises the 40 constituents of the CAC 40 index as of 2022/2023. We restrict our investment universe to ten stocks chosen across different sectors in order to favour diversification. We also prioritise companies with higher market capitalisation and ensure liquidity of stocks. Additionally, we maintain same-sector stocks (e.g. two luxury names, two aerospace names) to analyse their correlation and performance. The selected tickers and their sectors are listed in Table 1. The dataset spans 135 trading days, yielding 135 daily return observations per stock after computing first differences.

Table 1: Selected stocks and corresponding sectors.

Ticker	**Company**	**Sector**
MC	LVMH	Luxury Goods
RMS	Hermès	Luxury Goods
OR	L'Oréal	Consumer Goods
AIR	Airbus	Aerospace
SU	Schneider Electric	Industrials
SAF	Safran	Aerospace / Defence
TTE	TotalEnergies	Energy
BNP	BNP Paribas	Banking
AI	Air Liquide	Chemicals
SAN	Sanofi	Healthcare

2.2 Data Processing

For each stock the one-minute OHLC data is loaded from Excel files. The dataset is split into business days and the last observed close price of each day is retained as the daily closing price. All ten series are aligned on a common date range, truncated at the earliest final date across stocks. Daily simple (arithmetic) returns are then computed as

$$r_{i,j} = \frac{P_{i,j} - P_{i-1,j}}{P_{i-1,j}}, \tag{1}$$

where $P_{i,j}$ is the closing price of stock j on day i. The resulting returns dataframe covers the period from September 2022 to March 2023.

3 Portfolio Construction

3.1 Covariance and Correlation Matrices

The sample correlation and covariance matrices of the daily returns are estimated over the full sample. Figure 1 displays the correlation matrix. Most pairwise correlations are positive, which is typical for stocks belonging to the same national index. The luxury-goods names (MC, RMS, OR) exhibit particularly high mutual correlations (MC–RMS: 0.89, MC–OR: 0.74, RMS–OR: 0.76). In contrast, the healthcare stock SAN shows near-zero or slightly negative correlations with the rest of the universe (e.g. SAN–MC: -0.01, SAN–OR: -0.01), offering strong diversification benefits. The energy stock TTE also has relatively low correlations with most other names.

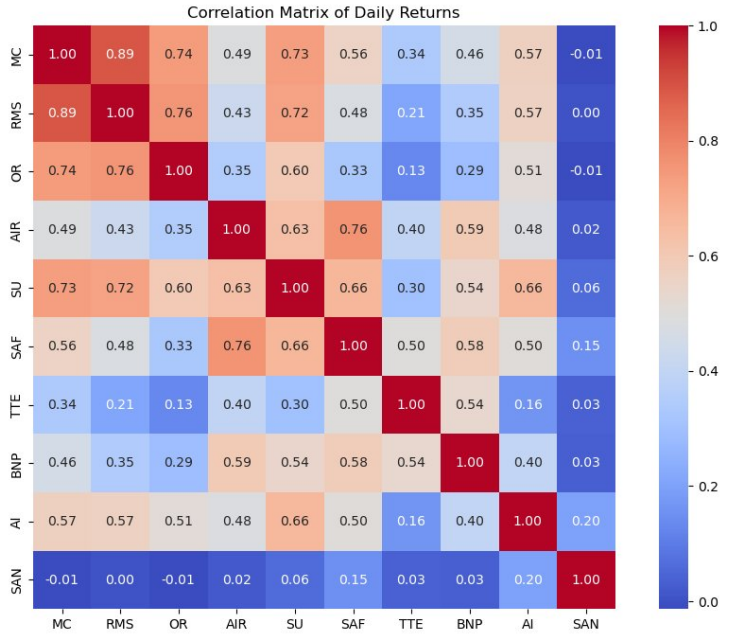


Figure 1: Correlation matrix of daily returns for the ten selected stocks.

The covariance matrix Σ (Figure 2) shows that BNP Paribas has the highest variance among the selected stocks (3.95×10^{-4}), followed by SU (3.39×10^{-4}) and RMS (3.39×10^{-4}). SAN exhibits the lowest variance, consistent with the defensive nature of the healthcare sector.

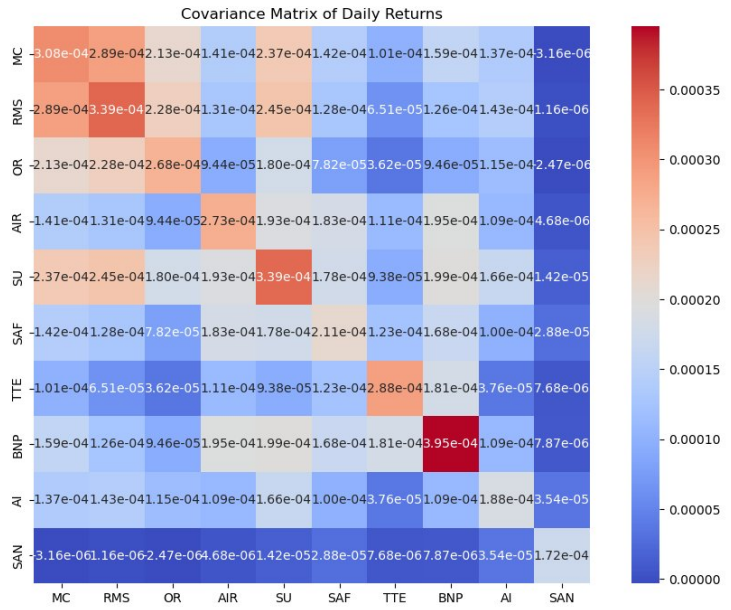


Figure 2: Covariance matrix of daily returns for the ten selected stocks.

3.2 Lag-1 Autocorrelation and Return Prediction

For each stock j we compute the lag-1 autocorrelation $\rho_j(1)$ of the daily return series. The results are reported in Table 2.

Table 2: Lag-1 autocorrelation of daily returns.

Stock	$\boldsymbol{\rho(1)}$
MC	-0.1936
RMS	-0.1747
OR	-0.1689
AIR	-0.1770
SU	-0.2327
SAF	-0.2504
TTE	-0.0828
BNP	-0.0128
AI	-0.1617
SAN	-0.0353

All autocorrelations are negative, indicating a mild mean-reversion pattern at the daily scale: a positive return on day i tends to be followed by a slightly negative return on day $i+1$, and vice versa. The magnitudes range from -0.01 (BNP) to -0.25 (SAF). However, according to the weak-form efficient market hypothesis, serial dependence in returns is typically weak, so any predictive signal derived from autocorrelation is expected to be limited.

The projected return at day $i+1$ for stock j is

$$R_{i+1,j} = \rho_j(1)\, r_{i,j}, \tag{2}$$

where $r_{i,j}$ is the observed return at day i.

Justification. Under an AR(1) model for each stock's return series, i.e. $r_{i+1,j} = \rho_j(1)\, r_{i,j} + \varepsilon_{i+1,j}$, the conditional expectation $\mathbb{E}[r_{i+1,j} \mid r_{i,j}]$ is exactly $\rho_j(1)\, r_{i,j}$. This is therefore the best linear predictor of the next-day return based solely on the current return and the lag-1 autocorrelation. Because all autocorrelations are negative, the model captures a short-term mean-reversion effect: after a positive day, it predicts a negative return, and conversely.

3.3 Markowitz Optimisation

Given the predicted return vector $R_{i+1} = (R_{i+1,1}, \ldots, R_{i+1,10})'$ from (2) and the covariance matrix Σ, the optimal weight vector for day $i+1$ is

$$\Omega_{i+1} = \frac{\Sigma^{-1} R_{i+1}}{\mathbf{1}_{10}' \, \Sigma^{-1} R_{i+1}}, \tag{3}$$

where $\mathbf{1}_{10}$ is a vector of ones of dimension 10. Given the predicted return vector and the covariance matrix Σ, the optimal portfolio weights for day $i+1$ are obtained from the Markowitz mean–variance framework. Under the full-investment constraint $\mathbf{1}'\omega = 1$, the Lagrangian approach yields the closed-form solution (3). This portfolio corresponds to the tangency portfolio when the risk-free rate is zero. Short selling is permitted (weights can be negative). In practice, extensions such as no-leverage constraints ($\sum_i |\omega_i| \leq 1$), sector bounds, transaction costs, or shrinkage estimators for Σ could significantly improve robustness.

The weights are recomputed daily: at the close of day i, the weights Ω_{i+1} are determined and applied to the realised returns of day $i+1$. The resulting weights are highly dynamic and can take extreme values. For example, on 2022-09-26 the weight on SU reaches -9.13 while SAF reaches $+6.87$, indicating very large leveraged positions. This is a well-known feature of unconstrained Markowitz portfolios: small changes in the estimated expected returns can lead to large swings in the optimal weights, especially when the covariance matrix is close to singular or the expected returns are small.

3.4 Realised Portfolio Returns

The realised portfolio return on day $i+1$ is

$$r_{i+1}^{\text{pf}} = \sum_{j=1}^{10} \Omega_{i+1,j}\, r_{i+1,j}. \tag{4}$$

Since no weights are available before day 2, the first portfolio return is set to zero. The first few realised daily portfolio returns are shown in Table 3.

Table 3: First five realised portfolio returns.

Date	$\boldsymbol{r}^{\text{pf}}$
2022-09-20	0.0000
2022-09-21	+0.0660
2022-09-22	−0.0782
2022-09-23	−0.0451
2022-09-26	+0.0737

The portfolio returns exhibit substantial day-to-day variation, reflecting the aggressive leveraged positions taken by the unconstrained Markowitz optimiser.

4 Risk and Performance Evaluation

4.1 Sharpe Ratios

The Sharpe ratio (Sharpe, 1966) measures risk-adjusted performance as the ratio of the expected risk premium over its standard deviation. Assuming a zero risk-free rate we compute two variants:

- **Ex-ante (predicted) Sharpe ratio**, based on the predicted portfolio returns:

$$\mathrm{SR}_{\mathrm{ante}} = \frac{\overline{\hat{r}^{\mathrm{pf}}}}{\hat{\sigma}(\hat{r}^{\mathrm{pf}})} = -0.0561.$$

- **Ex-post (realised) Sharpe ratio**, based on the actual portfolio returns:

$$\mathrm{SR}_{\mathrm{post}} = \frac{\overline{r^{\mathrm{pf}}}}{\hat{\sigma}(r^{\mathrm{pf}})} = 0.0296.$$

The ex-ante Sharpe ratio is slightly negative, reflecting the difficulty of predicting returns with weak autocorrelation signals. The ex-post Sharpe ratio is slightly positive, suggesting that the realised performance was marginally better than what the model predicted, though this is not meaningful given that it is only one realisation. Both values are very close to zero, indicating that the strategy does not deliver meaningful risk-adjusted returns. The difference between the two values is small and should not be over-interpreted, since it is based on a single realisation of the return process. This result is consistent with the weak autocorrelation structure of daily stock returns and with the weak-form efficient market hypothesis discussed in the course: as stock returns exhibit only small autocorrelations, they provide limited information for forecasting future returns.

As discussed in the course, the Sharpe ratio has well-known drawbacks: when negative it is hard to interpret (a manager could improve it by increasing risk), it treats upside and downside volatility symmetrically (a positive jump penalises the ratio), and it is sensitive to the time scale. This motivates the asymmetric measures below.

4.2 Upside Potential Ratio

The upside potential ratio (Sortino, van der Meer and Plantinga, 1999) compares the average positive return to the downside risk:

$$\mathrm{UPR} = \frac{\mathbb{E}\big[\max(r^{\mathrm{pf}}, 0)\big]}{\sqrt{\mathbb{E}\big[\min(r^{\mathrm{pf}}, 0)^2\big]}} = 0.3220. \tag{5}$$

The UPR belongs to the family of asymmetric risk-adjusted performance ratios discussed in the course, alongside the Sortino ratio and the Omega ratio. These metrics address the main shortcoming of the Sharpe ratio by focusing risk measurement on the lower partial moments rather than on total variance. More generally, the Kappa ratio K_n (van Harlow, 1991) unifies

these measures through n-th lower partial moments.

A UPR value below 1 indicates that the average gain on positive days is smaller than the downside semi-deviation, pointing to an unfavourable risk–reward profile.

4.3 Maximum Drawdown

The maximum drawdown (MDD) quantifies the largest peak-to-trough decline in cumulative portfolio value. Following the definition from the course, for a series of prices $P_1, \ldots, P_n$:

$$\text{MDD} = \max_{1 \le i < j \le n} \left(\frac{P_i - P_j}{P_i} \right). \tag{6}$$

In practice, this is computed efficiently by tracking, for each date j, the drawdown from the running peak: $m_j = 1 - P_j / \max_{1 \le i < j} P_i$, and taking $\text{MDD} = \max_j m_j$. Our portfolio yields $\text{MDD} = 1.4765$.

This value exceeds 1 (i.e. 100%), meaning the cumulative portfolio value went negative at some point during the sample due to the extreme leverage embedded in the unconstrained Markowitz weights. In a real leveraged portfolio this would correspond to a total wipeout.

4.4 Additive Volatility Contribution

Following the Euler decomposition presented in the course, the portfolio volatility $\sigma(R) = \sqrt{\sum_{j,k} \Omega_j \Omega_k \text{Cov}(r_j, r_k)}$ is homogeneous of degree 1 in the weights. By Euler's theorem, $\sigma(R) = \sum_i \Omega_i \frac{\partial \sigma(R)}{\partial \Omega_i}$, which yields the additive risk contribution of stock j on day i:

$$\text{RC}_{i,j} = \frac{\Omega_{i,j} \, (\Sigma \, \Omega_i)_j}{\sigma_{p,i}} = \frac{\Omega_{i,j} \, \sigma(r_j) \sum_k \Omega_{i,k} \, \rho(r_j, r_k) \, \sigma(r_k)}{\sigma_{p,i}}, \tag{7}$$

where $\rho(r_j, r_k)$ denotes the correlation between stocks j and k. By construction the contributions sum to the total portfolio volatility: $\sum_{j=1}^{10} \text{RC}_{i,j} = \sigma_{p,i}$.

This decomposition reveals which stocks are the main drivers of portfolio risk on each day. Stocks with large absolute weights (such as SU and SAF on leveraged days) or high covariance with the rest of the portfolio tend to have the largest contributions. On average across the sample, the stocks with the highest mean absolute risk contributions are SU, SAF and RMS, while TTE, SAN and BNP contribute the least, consistent with their lower weights and lower correlations with the rest of the portfolio. A numerical sanity check confirms that the row-wise sum of contributions equals the portfolio standard deviation to machine precision.

5 Discussion: Biases in Backtesting

Several methodological biases affect the quality of the backtest conducted above.

Look-ahead bias. This is the most critical issue in our analysis. Both the covariance matrix Σ and the lag-1 autocorrelations $\rho_j(1)$ were estimated over the *entire* sample period (September 2022 – March 2023) and subsequently used to construct and evaluate the portfolio on the *same* period. Information about the future was therefore implicitly available at every rebalancing date,

leading to in-sample overfitting. A more rigorous approach would use an expanding or rolling window: at each day i, estimate Σ and $\rho_j(1)$ only from data up to day i, then form the portfolio for day $i+1$.

Liquidity bias. The backtest assumes that the optimal weights can be executed at the observed closing prices without any market impact. In practice, each transaction entails costs (commissions, bid–ask spreads) and slippage, especially for large or leveraged positions. Furthermore, executing a trade itself modifies the price series through permanent and transitory impact, so the assumption that the portfolio can be rebalanced freely at displayed prices is unrealistic. This is particularly relevant here given the extreme weights produced by the unconstrained Markowitz optimiser.

Lack of statistical significance. Our backtest relies on a single realised trajectory of prices over roughly six months. A single trajectory is anecdotal and does not allow us to draw statistically robust conclusions. To strengthen the analysis, one could split the dataset into independent test sets (e.g. one per month) representing different market regimes, randomly draw the investment universe from the broader CAC 40 and check whether results are sensitive to stock selection, or apply statistical tests (such as a binomial test on the sign of daily returns) to validate the strategy. More generally, the average performance is less informative than the full conditional distribution of performance across scenarios.

References

- Markowitz, H. (1952). Portfolio Selection. *The Journal of Finance*, 7(1), 77–91.
- Course lecture notes — *Numerical Processing of Financial Data* (M. Garcin).